

BOPANA KUMAR SATYA SRI RAM

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SUMMARY

Software Development Engineer with 1+ years of full-time experience building production backend services, distributed systems, and real-time device infrastructure. Experienced in Python, FastAPI, Redis, Elasticsearch, REST APIs, event-driven architectures, and real-time streaming, with a focus on scalability, performance, observability, and production reliability.

SKILLS

Languages	Python, TypeScript, JavaScript, SQL
CS Fundamentals	Data Structures, Algorithms, Object-Oriented Design, Distributed Systems
Frameworks	FastAPI, React.js, Next.js
Backend	REST APIs, Microservices, Event-Driven Architecture
Databases	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch
Tools	Git, GitLab, Docker, Kong Gateway

EXPERIENCE

FlintLab Inc. (Jul 2025 - Present) *Software Development Engineer & Founding Engineer*

- Built backend services for a device testing platform supporting **100+ Android and 50+ iOS devices**, covering allocation, automation, streaming, recording, logging, and session lifecycle workflows.
- Developed Python/FastAPI services and REST APIs with Redis Pub/Sub for event-driven communication, supporting **20+ parallel test executions** across integration testing and production.
- Extended **ws-scrpy** with recording and platform integrations to support **50+ concurrent Android streams** with sub-second latency; built iOS streaming, device interaction, and recording services.
- Built the Android logging pipeline using allocation-triggered ADB Logcat collection, Redis Pub/Sub, Fluent Bit, and Elasticsearch; reduced representative log volume from **98K to 20K lines** and implemented retention controls to prevent production disk-pressure issues.
- Configured **Nginx reverse proxy + HTTPS/SSL via Certbot** to route separate development and production environments through dedicated subdomains behind a single static IP; implemented IPA resigning workflows for iOS execution.
- Implemented Auth0 authentication, RBAC, and API gateway token validation across platform services.

FlintLab Inc. (Mar 2025 - Jun 2025) *Software Development Engineer Intern*

- Reduced redundant frontend API calls by **30%** through state management and Redis caching; contributed to SPA migration that reduced observed routing time from **300 ms to 20 ms**.
- Migrated frontend API calls from **client-side to server-side execution** in Next.js, improving control over data access and reducing exposure of backend interactions to the client.
- Developed and integrated Appium-based automation workflows for **Android and iOS** devices, contributing to automated mobile application testing and device execution workflows.

PROJECTS

MyGVP Browser Extension & CLI – Built a Node.js browser extension and CLI used by 500+ students, processing 1,000+ academic result queries using MongoDB; distributed and versioned through the npm package ecosystem. [GitHub](#)

EDUCATION

Gayatri Vidya Parishad College of Engineering (GVPCE) , Visakhapatnam	2021 - 2025 Bachelor of Technology in Computer Science and Technology
	CGPA: 8.0